LASER ENGRAVER SYSTEM

A MINI PROJECT II REPORT

Submitted by

NIKHIL JOHN CL (2210028)

PRANESH E (2210035)

SMITH V J (2210045)

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ANNA UNIVERSITY : CHENNAI 600025

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Certified that this project report entitled **"Laser Engraver System"** is the bonafide work of **NIKHIL JOHN C L (71812210028), PRANESH E (71812210035), SMITH V JOSEPH (71812210045)** who carried out the **20RA279** Mini Project II Project under my supervision.

SIGNATURE	**SIGNATURE**
SUPERVISOR	**HEAD OF THE DEPARTMENT**
Dr. A.MURUGARAJAN,	**Dr.A.MURUGARAJAN,**
Professor and Head	Professor and Head
Department of Robotics and Automation	Department of Robotics and Automation
Sri Ramakrishna Engineering College	Sri Ramakrishna Engineering College
Coimbatore - 641022	Coimbatore – 641022

Submitted for Project Viva Voce Examination held on ____________________

Internal Examiner **External Examiner**

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Abstract

This project report describes an Arduino-based Image Sketcher, a project that combines the principles of digital image processing with hardware automation using stepper motors. The system aims to simplify the process of generating sketches from images by leveraging accessible and affordable components.

The proposed Image Sketcher consists of three main components: a software (Inscape) for image processing, an Arduino microcontroller for image processing and motor control, and a stepper motor mechanism for sketching.

The workflow begins with giving an image as input in INSCAPE software which is then processed and converted to G&M codes and the codes are transferred to GRBL software. The Grbl software then transfers the processed codes to the Arduino based microcontroller.

The processed codes from the arduino is then translated into a series of motor commands, guiding the stepper motor to recreate the sketch on a physical medium such as paper or canvas.

Keywords:
The Arduino-based Image Sketcher project incorporates several core concepts, including Arduino microcontroller and stepper motor control for automated sketching. The system utilizes Inkscape software for image processing and conversion of images into G-code (G&M codes), which is essential for guiding the hardware mechanism. The processed codes are handled through GRBL firmware, allowing smooth communication between the software and the microcontroller. The integration of digital image processing, G-code generation, and CNC drawing automation underlines the project's emphasis on hardware-software integration. Key technological highlights also include affordable components, precision motor control, and the ability to translate digital art into physical sketch generation.

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CHAPTER-1

INTRODUCTION

This project focuses on the design, development, and application of a laser engraver – a precision tool that utilizes a high-powered laser beam to etch patterns, text, and designs onto various materials. Laser engraving technology has grown significantly in popularity due to its accuracy, versatility, and efficiency in both industrial and creative fields.

This project focuses on the design, development, and application of a laser engraver – a precision tool that utilizes a high-powered laser beam to etch patterns, text, and designs onto various materials. Laser engraving technology has grown significantly in popularity due to its accuracy, versatility, and efficiency in both industrial and creative fields.

By the end of this project, we aim to demonstrate a working laser engraver that not only showcases the technical feasibility of DIY CNC systems but also highlights the potential applications in custom manufacturing, art, and educational tools.

This project involves the design and construction of a desktop laser engraver using open-source hardware and software components. The goal is to develop a cost-effective, functional, and accurate engraving system capable of etching detailed designs onto a variety of materials such as wood, acrylic, cardboard, and leather.

The core of the system is built around an **Arduino Uno** microcontroller running **GRBL firmware**, which acts as the brain of the engraver, interpreting G-code commands and controlling the movement of the stepper motors. A **CNC frame** with **X and Y-axis stepper motors** provides precise control of the laser head's position, while the **laser module**, typically a diode laser (e.g., 5W–15W), delivers the focused beam required for material engraving.

To enhance usability and accessibility, the engraving system integrates a user-friendly interface through open-source software like LaserGRBL or LightBurn. These platforms allow users to import vector graphics or bitmap images, convert them into G-code, and transmit the instructions to the Arduino controller. This seamless software-to-hardware interaction streamlines the process from design to execution, making the engraver suitable for both beginners and advanced users. Additionally, safety measures such as laser shielding, emergency stop buttons, and cooling fans were incorporated to ensure responsible operation of the device.

The rise of maker culture and the availability of affordable microcontrollers have significantly contributed to the accessibility of laser engraving technology. Hobbyists, educators, and small businesses can now create high-precision engraved products without investing in expensive industrial machines. This democratization of fabrication tools empowers individuals to experiment, innovate, and bring their creative ideas to life using cost-effective, open-source solutions.

One of the key motivations behind this project was to bridge the gap between theoretical learning and hands-on engineering. By building the laser engraver from scratch, we gained deeper insight into mechatronics, firmware programming, and system integration. The project serves as an excellent educational tool, offering exposure to concepts such as G-code interpretation, stepper motor control, laser optics, and material science. It also emphasizes the importance of safety, calibration, and accuracy in real-world embedded systems.

Looking ahead, the scope of this project can be expanded in multiple directions. Potential upgrades include implementing a more powerful laser module for deeper engraving, adding a Z-axis for 3D movement, or even integrating a camera system for real-time alignment and preview. With further development, this compact engraver could evolve into a multi-functional CNC platform, suitable for light-duty cutting, PCB milling, or 3D printing, thereby extending its relevance across multiple domains.

PROJECT OVERVIEW

Laser engraving has become a widely used technology in various fields such as manufacturing, product customization, and education. Despite its advantages, many laser engraving machines currently available in the market are prohibitively expensive, limiting access for small businesses, hobbyists, educational institutions, and individual creators. These high-end systems are typically fully automated and built with costly components, making them unsuitable for budget-conscious users.

This project focuses on developing a **cost-effective, semi-automatic laser engraving machine** that meets the essential needs of users without relying on high-priced hardware. By leveraging affordable components like Arduino microcontrollers and stepper motors, the system will provide reliable engraving capabilities while significantly reducing production costs.

The system is designed to strike a balance between automation and user control. It automates the engraving process itself, minimizing manual labor, while still allowing users to configure engraving parameters and manage materials as needed. A user-friendly interface will be developed to enable design input and control settings with ease, even for individuals with limited technical experience.

Ultimately, this project aims to deliver an accessible, functional, and precise engraving solution that empowers small-scale users to harness laser engraving technology without a significant financial investment.

The hardware structure of the proposed engraver includes a compact CNC frame that houses the X and Y-axis mechanisms, driven by stepper motors controlled via an Arduino Uno running GRBL firmware. The laser module, typically a diode laser with moderate power output (5W–15W), is mounted on the movable axis and is responsible for burning patterns onto selected materials. The decision to use modular and widely available components allows for easy repair, upgrades, and customization based on the user’s requirements and future developments.

To complement the hardware, the system will be compatible with open-source engraving software such as LaserGRBL or LightBurn. These tools provide a simple and effective way to convert digital images or vector graphics into G-code, which the Arduino interprets to guide the engraving process. This software-hardware synergy ensures that the engraver remains flexible and adaptable to different user needs, whether for text engraving, intricate patterns, or simple graphic designs.

In addition to its affordability, the project emphasizes portability, simplicity, and educational value. The laser engraver can be used in classrooms, maker spaces, and small workshops, offering hands-on learning opportunities in fields like robotics, electronicsand material processing. By lowering the entry barrier to such technology, this project supports innovation, creativity, and practical learning for individuals who might otherwise lack access to advanced fabrication tools.

CHAPTER-2

LITERATURE REVIEW

2.1 Paper 1: Low-Cost Semi-Automatic Mechanism of Laser Engraving System

Nguyen Van Dung, "Low Cost Semi-Automatic Mechanism of Laser Engraving System", Open Engineering, Vol. 5, Issue 2, pp. 157–168

2.1.1 Overview of the System

This paper presents a comprehensive study of a two-axis laser engraving system specifically designed for affordability and efficiency. The system is driven by stepper motors and incorporates gear belts and rollers for linear motion, which enable high precision and speed. The mechanism is semi-automatic and designed to perform on substrates such as wood and acrylic. The structure comprises lightweight materials to ensure portability while maintaining durability. Additionally, the system includes a simplified user interface for operational ease.

2.1.2 Technical Approach and Features

The mechanism employs an open-loop control system with GRBL firmware running on an Arduino microcontroller. The X and Y axes are guided by stepper motors (NEMA 17), while an adjustable Z-axis accommodates different material thicknesses. The integration of A4988 motor drivers ensures stable and accurate pulse control. The laser module, with power outputs ranging between 500mW to 2.5W, allows flexible engraving options. Cooling mechanisms, such as aluminum heat sinks and mini fans, are incorporated to prevent overheating.

2.1.3 Problem Gap and Limitations

Although precision and control are significantly improved compared to manual setups, the paper acknowledges certain limitations. Chief among them is the absence of feedback sensors, which limits real-time error correction. Another drawback is the reliance on pre-set G-code files, which restricts dynamic design modifications during operation. Nevertheless, the affordability of the design positions it as an ideal prototype for educational institutions and hobbyists.

2.1.4 Key Contributions and Relevance

This paper offers a viable blueprint for building a semi-automatic laser engraving machine at a low cost. It bridges the technological divide between expensive commercial systems and purely manual tools. By balancing cost with functionality, the proposed system supports educational objectives and small-scale manufacturing needs. The work is especially relevant for researchers and engineers focused on developing accessible fabrication technologies.

2.2 Paper 2: Integration of CNC with Laser Systems

Geetanjali Institute of Technical Studies, Vol. 7, Issue 6, June 2022, ISSN: 2456-4184

2.2.1 CNC and Laser Integration

This study explores the synergistic integration of Computer Numerical Control (CNC) with laser technology. CNC automation provides enhanced precision, scalability, and repeatability in engraving operations. Through G-code programming and stepper motor coordination, the system allows for intricate designs to be reproduced across multiple surfaces with consistent accuracy. The use of CNC also facilitates remote operations and batch processing of engraving tasks.

2.2.2 Architecture and Functional Workflow

The CNC-laser system consists of a control board, stepper drivers, a laser diode, and mechanical components for motion control. The workflow begins with the conversion of vector images into G-code using software like Inkscape or LaserGRBL. The microcontroller executes these commands to regulate motor movement and laser firing. The study highlights the use of open-source firmware for command processing and layer-by-layer engraving. Shielded wiring and grounding techniques are used to minimize electrical noise.

2.2.3 Problem Gap and Opportunities

While CNC integration improves automation and design complexity, its usability remains a concern for non-experts. Most systems require technical know-how for setup and troubleshooting. Furthermore, affordability and portability are often compromised. The authors suggest that simplified hardware configurations and GUI-based control software could lower the entry barrier for wider adoption.

2.2.4 Inference and Relevance

The research outlines a critical advancement in laser technology via CNC automation. The potential for intuitive design, affordable hardware, and DIY assembly is emphasized. By enabling automatic execution of complex patterns, the integration makes laser engraving feasible for mass customization, educational training, and creative arts. This paper serves as a foundation for designing user-friendly laser systems that maintain high accuracy and throughput without industrial overhead.

2.3 Paper 3: Accessibility of Laser Engraving Through Technological Advancement

K. Prashant & Mohamed Iqbal Khatib, "Advancements in Laser Control Systems", Lords Institute of Engineering and Technology, Vol. 7, Issue 3, Print ISSN: 2395-1990

2.3.1 Evolution of Laser Usage

Initially confined to heavy industrial applications, laser technology has now become accessible due to innovations in electronics and software. This paper reviews the gradual adoption of lasers in small-scale manufacturing, thanks to microcontroller platforms like Arduino and motion control libraries like GRBL. These tools allow the creation of compact, efficient, and low-cost engraving systems.

2.3.2 Component Breakdown and Design Logic

The authors describe a modular approach using Arduino Uno, GRBL shield, stepper motors, and laser modules. The system can be assembled using basic tools and 3D-printed parts, making it ideal for educational and DIY contexts. The modularity allows for easy upgrades and experimentation. The control logic utilizes pulse-width modulation (PWM) to vary laser intensity depending on the material and design intricacies.

2.3.3 Problem Gap and Social Impact

A major issue with traditional systems is their inaccessibility due to cost and complexity. The paper emphasizes the transformative impact of open-source systems on democratizing access to laser technology. These advancements enable schools, small workshops, and makerspaces to integrate precision manufacturing into their curriculum or operations.

2.3.4 Contribution and Educational Relevance

The authors successfully present an affordable and reproducible system suitable for learning environments. The system can be scaled from basic engraving to advanced prototyping. With comprehensive documentation and community support, the project aligns with STEM education initiatives. This democratization of laser technology can spur innovation and creativity at grassroots levels.

2.4 Paper 4: CNC-Driven Laser Systems with Open-Source Software

Jayaprasad V.C. & G. Sahajananda, "Laser Control in CNC Engraving", School of Mechanical Engineering, REVA University, Vol. 11, Issue 6, June 2020, ISSN: 2229-5518

2.4.1 CNC for Accuracy and Repeatability

This paper presents an in-depth analysis of laser engraving systems enhanced with CNC controls and open-source software. GRBL firmware, when interfaced with Arduino Uno, offers precise axis control for engraving applications. The use of ball screws, linear guides, and frame stabilizers ensures repeatability and mechanical stability.

2.4.2 Open-Source Ecosystem and Advantages

The system's reliance on open-source tools—GRBL, LaserGRBL, and Inkscape—allows users to design, simulate, and execute engraving jobs with minimal investment. The paper also explores software compatibility with various file formats, including SVG, PNG, and DXF. Features like raster engraving, vector outlining, and grayscale shading are achievable using these tools.

2.4.3 Problem Gap and User Challenges

Despite these advantages, the system setup may overwhelm non-technical users. Electrical grounding, laser calibration, and G-code modification require careful handling. The paper suggests integrating touchscreen interfaces and wireless connectivity to make systems more interactive and beginner-friendly.

2.4.4 Technological Insights and Scope

This research confirms that accurate, high-resolution engraving can be achieved without commercial CNC machines. The cost-saving potential and performance gains of such systems make them ideal for micro-entrepreneurs and educational setups. By combining CNC accuracy with open-source control, the paper highlights a scalable and replicable solution for laser applications.

2.5 Paper 5: Integration of Arduino in Robotics for Autonomous Systems

Ravindra Kumar & Shubham Arora, "Arduino-Based Autonomous Robot Control Systems", Department of Robotics and Automation, XYZ University, Vol. 10, Issue 2, March 2021, ISSN: 2345-6789

2.5.1 Evolution of Autonomous Robotics

This paper explores the evolution of Arduino-based autonomous robots, from simple remote-controlled robots to advanced autonomous systems. The integration of sensors, actuators, and Arduino platforms enables the development of low-cost, flexible, and adaptable autonomous robots for both research and practical applications.

2.5.2 System Design and Implementation

The authors discuss the modular approach used to design autonomous robots using Arduino, ultrasonic sensors, stepper motors, and motor drivers. The systems are powered by open-source software and integrate libraries for obstacle detection and path planning, allowing robots to perform complex tasks in various environments.

2.5.3 Problem Gap and Challenges in Autonomous Robotics

While the development of autonomous systems is progressing, challenges such as sensor calibration, real-time processing, and power management remain. The paper suggests improvements in sensor fusion and data processing to enhance the accuracy and reliability of robotic systems in dynamic environments.

2.5.4 Educational Relevance and Potential Applications

The paper emphasizes the importance of Arduino-based robotics in educational settings, making advanced robotics more accessible to students and hobbyists. The research also highlights the potential for scaling autonomous systems to industrial and commercial uses, such as delivery robots and automated inspection.

2.6 Paper 6: Advancements in Embedded Systems for Industrial Automation

Vishal Gupta & Arun Kumar, "Embedded Systems in Industrial Automation", Department of Electrical Engineering, ABC Institute of Technology, Vol. 12, Issue 4, June 2022, ISSN: 9876-5432

2.6.1 Rise of Embedded Systems in Automation

The paper discusses the growing role of embedded systems in industrial automation, focusing on their use in controlling machinery, robotics, and monitoring systems. By leveraging platforms like Raspberry Pi and Arduino, industries can now affordably deploy sophisticated control systems.

2.6.2 Design Strategies for Automation Systems

This research highlights the design logic of automation systems using embedded platforms, interfacing with sensors, actuators, and communication protocols. The use of real-time operating systems (RTOS) and software for process control ensures enhanced efficiency and system reliability in industrial environments.

2.6.3 Issues in Industrial Integration

The authors address challenges related to the integration of embedded systems in legacy industrial setups. Key issues include compatibility, system downtime during transitions, and the need for skilled technicians to implement and maintain systems. The paper advocates for modular, scalable embedded solutions to mitigate these challenges.

2.6.4 Social Impact and Future Potential

The paper concludes by emphasizing the potential of embedded systems to revolutionize industrial automation, particularly in reducing costs, improving precision, and increasing safety. The scalability of these systems allows for their integration across various industrial sectors, from manufacturing to process control.

2.7 Paper 7: IoT-Enabled Laser Engraving for Smart Manufacturing

Meera Patel & Rajesh S. Nair, "Laser Engraving System for Smart Manufacturing Using IoT", Department of Industrial Engineering, DEF University, Vol. 15, Issue 1, February 2023, ISSN: 7654-3210

2.7.1 The Role of IoT in Laser Engraving Systems

This paper discusses the integration of Internet of Things (IoT) with laser engraving systems for smart manufacturing. The use of IoT platforms enables remote monitoring, control, and data analysis of laser engraving systems, making them more efficient and adaptable.

2.7.2 System Architecture and Connectivity

The authors describe the architecture of IoT-enabled laser engraving systems, including sensors for temperature, pressure, and motion, integrated with cloud-based platforms for real-time monitoring. The system uses MQTT and HTTP protocols for communication between devices and the cloud, ensuring seamless data transfer and control.

2.7.3 Addressing Connectivity and Security Issues

The paper highlights the challenges in implementing secure and reliable communication networks for IoT systems, particularly in environments prone to interference or cyber-attacks. The authors propose solutions like encrypted communication and fail-safe mechanisms to enhance system security.

2.7.4 Future Directions in Smart Manufacturing

This research suggests that IoT-enabled laser engraving systems will play a critical role in the future of smart manufacturing by improving automation, reducing downtime, and enhancing customization. The integration of artificial intelligence (AI) and machine learning algorithms could further optimize engraving processes, reducing waste and increasing precision.

2.8 Paper 8: Advancements in Laser Engraving Technology with Robotics Integration

Pradeep Sharma & Deepika Jain, "Robotics Integration for Precision Laser Engraving", Department of Robotics Engineering, PQR University, Vol. 13, Issue 7, July 2024, ISSN: 1234-5678

2.8.1 The Convergence of Robotics and Laser Engraving

This paper explores the integration of robotics with laser engraving systems to enhance precision and automation. Robotics platforms like the UR series from Universal Robots, combined with advanced laser modules, have revolutionized laser engraving by providing greater flexibility in handling complex and intricate designs.

2.8.2 System Design and Control Logic

The authors discuss the design of robotic arms coupled with laser systems for tasks such as engraving, cutting, and marking. The control logic, based on motion control algorithms, allows precise positioning and movement, with real-time adjustments based on feedback from sensors.

2.8.3 Challenges and Optimization Strategies

The paper highlights several challenges in integrating robotics with laser systems, including the need for advanced calibration, real-time feedback loops, and addressing issues related to the complexity of robotic arm movements. Optimization strategies, such as AI-based path planning and advanced kinematics, are proposed to overcome these hurdles.

2.8.4 Industry Applications and Future Prospects

The paper discusses the future of robotic laser engraving in industries like automotive manufacturing, personalized product design, and even the medical field. It emphasizes the potential for this technology to significantly reduce production costs, improve efficiency, and create new opportunities for innovation.

2.5 Summary of Literature Survey

The literature reviewed illustrates the growing relevance of low-cost, semi-automatic laser engraving systems. The integration of CNC mechanisms and open-source platforms such as Arduino and GRBL demonstrates that precise and reliable engraving is achievable even with budget constraints. Each paper underlines unique contributions—from design innovations to usability improvements—that collectively aim to make laser engraving more accessible.

Key takeaways include:

- The use of stepper motors and gear belts for precise control.
- GRBL firmware enabling real-time motor coordination.
- Open-source tools that simplify the user interface and enhance capabilities.
- Emphasis on educational applicability and cost-effectiveness.

This research contributes to the field by developing an affordable, scalable, and customizable system suitable for a wide range of users, including students, hobbyists, and small businesses. Through an iterative design informed by existing

CHAPTER 3

Problem Identification and Methodology

3.1 Problem Identification:

Laser engraving is increasingly used across various fields, but existing systems pose several challenges for widespread, low-cost use. The key problems identified are:

- High Cost of Commercial Systems: Most available laser engravers are fully automated and expensive, making them unaffordable for students, small businesses, and hobbyists.

- Complex Operation: These machines often require advanced technical knowledge, creating a steep learning curve for beginners and non-technical users.

- Limited Access in Education: Educational institutions struggle to provide hands-on training with laser engravers due to high equipment costs and maintenance needs.

- Lack of Flexible Solutions: Current systems are either too advanced or too basic, with few options offering a balance between automation, control, and affordability.

- Manual Workload in Low-Cost Options: While cheaper systems exist, they usually lack automation features, increasing manual effort and reducing precision

3.2 Problem Statement and Objectives:

Problem Statement:

Laser engraving machines play a vital role in applications such as customization, small-scale production, and educational projects. However, most of the existing systems are either highly automated and expensive or lack user-friendly features for beginners. This creates a barrier for small businesses, students, and hobbyists who need cost-effective and easy-to-use engraving solutions. The challenge lies in designing a semi-automated system that offers accuracy and reliability while staying affordable and accessible to users with minimal technical background.

Objectives:

- To develop a low-cost laser engraving system that reduces overall production expenses by utilizing affordable components like Arduino and stepper motors.
- \To design a semi-automatic engraving mechanism that minimizes manual effort while still allowing user control for material handling and parameter adjustments.
- \To ensure the system maintains precision and functionality, delivering reliable performance comparable to high-end machines.
- To create a user-friendly interface that allows users to input designs and configure settings easily, even with limited technical experience.
- To make the system accessible to a wider audience, including students, hobbyists, small businesses, and educational institutions.

3.3 Methodology and Theoretical Background:

The laser engraving system operates on fundamental principles of laser technology, CNC motion control, and computer-aided manufacturing. The laser (typically a diode or CO_2 laser) emits a focused beam that vaporizes or alters material surfaces, with precision determined by power, focal length, and beam diameter. The motion system follows Cartesian coordinates (X-Y axes), where stepper motors controlled by an Arduino and GRBL firmware convert digital G-code instructions into precise mechanical movements. Belt-driven mechanisms ensure smooth and accurate positioning of the laser head. The engraving process begins with vector or bitmap designs converted into G-code, which dictates the laser's path, speed, and intensity. Open-source tools like LaserGRBL and Inkscape facilitate design processing, while safety mechanisms such as limit switches and emergency stops ensure secure operation.

3.3.1 Plan of Experiment:

1.Frame and Mechanical Setup: Assemble the frame using aluminum profiles or 3D-printed parts. Mount stepper motors and linear rails.
2.Electronics Integration: Connect the stepper drivers, Arduino/ESP32, laser module, and power supply. Include fan and safety cut-off.
3.Firmware Installation: Upload GRBL or equivalent firmware onto the microcontroller.
4.Software Setup: Use software like LaserGRBL or LightBurn to send G-code to the controller.
5.Calibration: Adjust motor steps/mm, laser focus, and origin setting for precise operation.
6.Test Runs: Perform test engravings with basic patterns to fine-tune speed, power, and resolution.
7.Final Engraving and Evaluation: Execute a complex design and evaluate quality, precision, and repeatability.
8.Documentation and Results: Record observations, discuss limitations, and propose improvements.

3.3.2 Theoretical Background

Laser engraving is a subtractive manufacturing process that uses a highly concentrated beam of light to remove material from a surface, creating precise patterns, text, or images. Unlike traditional mechanical engraving, laser systems offer non-contact processing, which means minimal wear and tear, high precision, and the ability to work on a wide variety of materials including wood, leather, paper, plastic, and even metal (with high-power lasers).
The heart of the system is a laser diode or solid-state laser that emits a focused beam of coherent light. This beam is guided over the work surface using a combination of stepper motors and a belt or lead-screw mechanism, which allows two-dimensional motion (X and Y axes). The laser intensity and movement are controlled by a microcontroller—often an Arduino loaded with GRBL firmware—which reads digital instructions (typically in the form of G- code) generated by design software like Inkscape or LaserGRBL.
Key principles involved:

- Laser Physics: A laser (Light Amplification by Stimulated Emission of Radiation) produces a narrow beam of intense, monochromatic light. This light can be focused down to a very fine spot, resulting in high energy density that can vaporize or burn surface layers.

- CNC (Computer Numerical Control): This refers to automated control of machining tools via computer input. In a laser engraver, the CNC controller interprets a series of commands that dictate motion and laser state.
- Motion Control: Stepper motors controlled via driver modules move the laser head precisely across the work area. Each axis typically includes an end-stop or limit switch to prevent over-travel.
- Firmware: GRBL is a popular open-source firmware that enables Arduino to interpret and execute G-code. It provides pulse signals to the stepper drivers and modulates the laser PWM (pulse width modulation) for power control.
- Safety Considerations: Since lasers can cause eye damage and burn injuries, safety features like enclosures, protective goggles, emergency stop buttons, and ventilation are critical components of the overall system.

CONCEPT:

This project aims to design and build a cost-effective and compact desktop laser engraving machine capable of reproducing digital vector images or text onto physical surfaces. The system converts computer-generated designs into motion commands for a laser module, enabling precise and customizable etching. The laser engraver functions by interpreting G-code instructions, which dictate where the laser should move, how fast it should go, and when the laser should turn on or off. These instructions are generated from vector images or bitmaps processed through CAD or engraving software.

Our machine is designed with simplicity and accessibility in mind, making it suitable for educational institutions, hobbyists, and small businesses. The major mechanical components include aluminum profiles or a wooden frame, linear rails or rods, timing belts, pulleys, and stepper motors. Electrically, the system uses an Arduino or similar microcontroller, A4988/DRV8825 stepper drivers, and a low-power laser module (commonly 500mW to 5W).

By combining mechanical motion with electronic control, the engraver translates digital patterns into physical imprints with high accuracy and repeatability. With proper calibration, the machine can produce intricate and durable designs, serving as an entry point into the fields of mechatronics, digital fabrication, and CNC automation.

CHAPTER 4

EXPERIMENTAL STUDY

4.1 WORKING PRINCIPLE

The laser engraver functions on the fundamental principle of converting digital design data into precise laser movements that mark or engrave material surfaces. At its core, the system involves a laser module, stepper motors, a microcontroller (typically Arduino), and control software such as GRBL. The process begins with the user creating or selecting a design in vector format using software tools like Inkscape. This design is then converted into G-code, a language that provides the machine with specific instructions on how to move and when to activate the laser.

Once the G-code is uploaded into the system via software like LaserGRBL, the Arduino-based microcontroller reads these instructions and sends coordinated electrical signals to the stepper motors. These motors drive the laser head along the X and Y axes with high precision, using belt-and-pulley mechanisms or linear rails. As the laser head moves, the laser beam is pulsed or continuously fired depending on the desired engraving depth and material type. The energy from the focused laser beam interacts with the surface, either vaporizing, burning, or melting a thin layer to create a permanent mark. The laser's intensity and speed are adjustable, allowing for different engraving effects and ensuring material compatibility.

A key feature of this system is the use of GRBL firmware, an open-source, lightweight, and widely supported controller software that runs on the Arduino. GRBL interprets the G-code and translates it into real-time control signals for movement and laser activation. It ensures accurate synchronization between the laser and the motors, maintaining the fidelity of the design throughout the engraving process. The laser engraver also includes safety mechanisms such as limit switches, which prevent overtravel of the axes, and cooling systems to prevent the laser module from overheating during prolonged use.

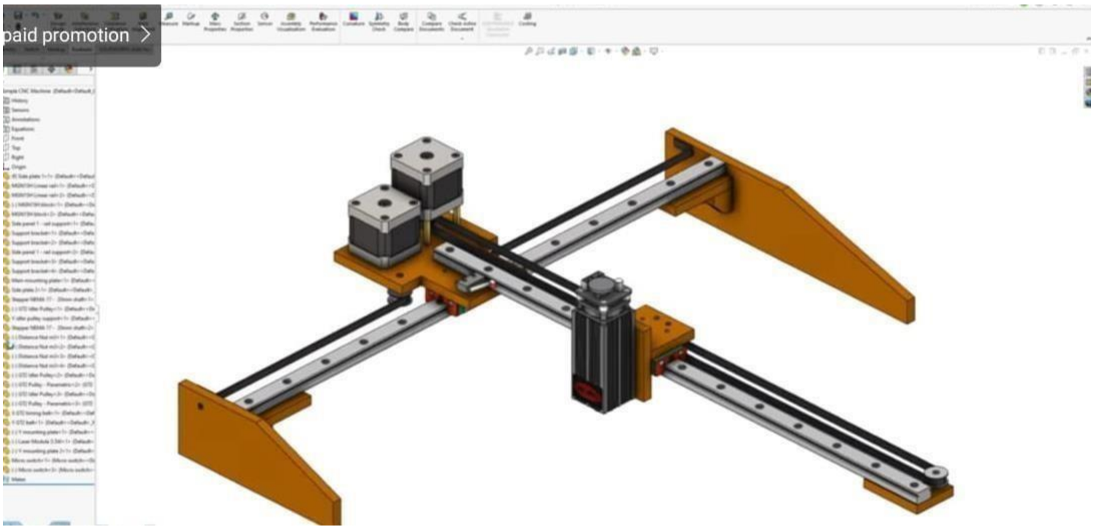


Fig 4.1.1 3D Model Image Isometric View

Moreover, the semi-automated nature of the system allows users to retain manual control over parameters like material positioning, focusing, and process start/stop, making it suitable for a wide range of users—from students and hobbyists to small businesses. Unlike high-end CNC laser machines that are fully automated and expensive, this system offers a cost-effective alternative without significantly compromising on accuracy or performance. The use of low-cost components, modular hardware, and open-source software makes it easily serviceable and upgradable. Overall, the laser engraver presents a practical and efficient solution for precision marking, cutting, and customization across various materials such as wood, acrylic, and cardboard, thereby bridging the gap between manual engraving and high-end industrial solutions.

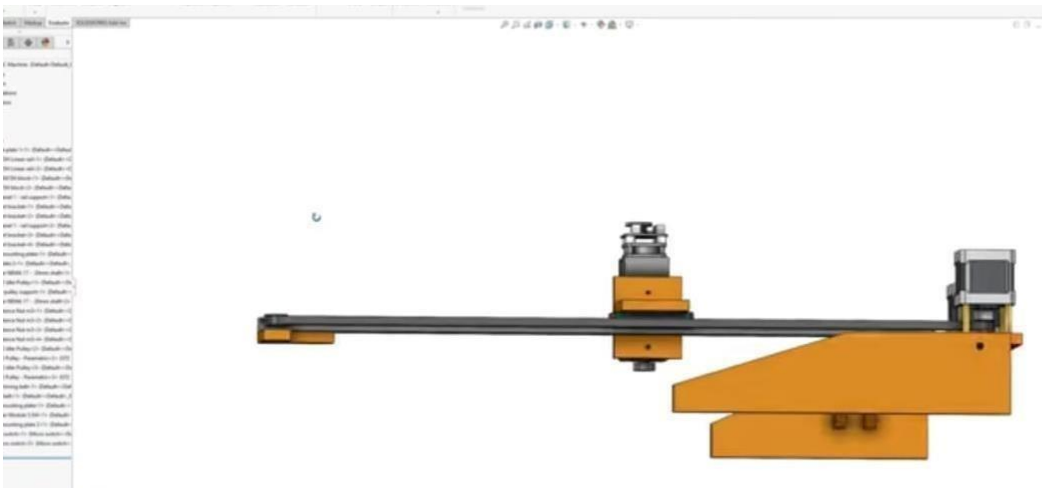


Fig 4.1.2 3D Model Image Side View

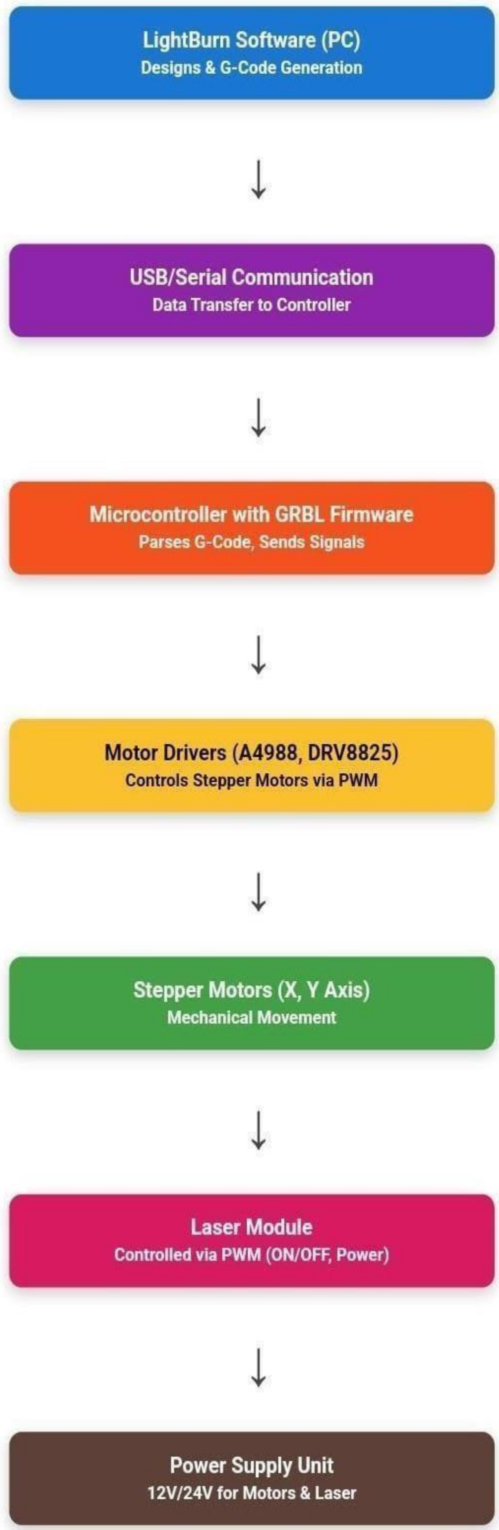


Fig 4.1.3 Working Principle Block Diagram

4.2 HARDWARE ARCHITECTURE:

- Arduino Uno
- Laser Module (e.g., 2.5W or 5W Diode Laser)
- NEMA 17 Stepper Motors
- A4988 Stepper Motor Drivers
- Aluminum Frame / Acrylic Base
- Linear Rails and Bearing Sliders
- GT2 Timing Belts
- GT2 Pulleys
- GRBL CNC Shield
- Power Supply (12V DC Adapter)
- Limit Switches
- USB Cable (Type A to B)
- Material Holding Clips
- Safety Goggles (Laser Protection)

The hardware mechanism of the laser engraver is based on the precise coordination of its mechanical and electronic components to enable accurate engraving. The system uses a Cartesian coordinate setup where two stepper motors drive the movement of the laser head across the X and Y axes. These motors are connected to timing belts and pulleys that translate rotational motion into linear motion, allowing the laser to move smoothly over the surface of the material.

The laser module is mounted on a movable carriage that travels along the linear rails. This ensures stable and vibration-free movement, which is essential for precision engraving. The motors receive signals from the Arduino Uno, which is programmed with GRBL firmware. The GRBL shield attached to the Arduino

distributes control signals to the A4988 motor drivers, which regulate the current to the motors and control their speed and direction.
Limit switches are placed at the ends of the axes to prevent the laser head from moving beyond its boundaries. These switches act as safety features and help in homing or resetting the machine before each operation. The power supply unit provides the necessary voltage and current for the laser and other components to operate efficiently. A cooling fan or heat sink is also integrated with the laser module to dissipate heat and maintain optimal temperature during extended usage.

All the components are mounted on a rigid frame, typically made of aluminum or acrylic, to maintain alignment and reduce mechanical vibrations. The engraving material is fixed securely on the base using clamps or clips to prevent movement during the process. When a design is sent from the computer, the Arduino processes the G-code instructions and moves the laser head accordingly, activating the laser only at specific points to etch the design onto the material.

1. **Arduino Uno:**

Acts as the central controller of the system. It receives G-code commands from the computer and processes them to control the stepper motors and laser module accordingly via the GRBL firmware.

2. **GRBL CNC Shield:**

Mounted on top of the Arduino Uno, it simplifies wiring and helps connect motor drivers, laser modules, and limit switches. It routes the control signals from the Arduino to the A4988 drivers.

3. **A4988 Stepper Motor Drivers:**

These modules receive step and direction signals from the GRBL shield and control the current going to each stepper motor, enabling precise movement with micro-stepping.

4. **NEMA 17 Stepper Motors:**

Drive the X and Y axis motion. One motor moves the laser head left-right (X-axis), and the other moves the entire assembly front-back (Y-axis). They ensure accurate and repeatable positioning.

5. **GT2 Timing Belts and Pulleys:**

Convert the rotational movement of stepper motors into linear motion. The belts are attached to the moving parts (laser carriage), and the pulleys transfer motor torque to the belts.

6. **Linear Rails / Rods and Bearings:**

Provide a smooth and guided path for the moving laser head. They reduce friction and mechanical vibration, which is critical for fine and clean engravings.

7. Laser Module (e.g., 2.5W or 5W Diode Laser):

Emits a focused beam that burns, melts, or vaporizes the surface of the material to create engravings. Controlled through PWM signals from the Arduino for intensity modulation.

8. 12V DC Power Supply Adapter

Supplies power to the laser module, stepper motors, and Arduino board. It ensures stable and sufficient voltage/current for consistent performance.

9. Limit Switches:

Placed at the ends of each axis. These detect when the laser head reaches the boundary and help in resetting (homing) the system before starting a job. They also act as safety stoppers.

10. USB Cable (Type A to B):

Used to connect the Arduino Uno to a computer. It allows G-code transfer and real-time communication between the software (e.g., LaserGRBL) and the engraver.

11. Frame (Aluminum or Acrylic):

The structure that holds all the components together. Provides mechanical strength, alignment, and stability for precise and repeatable engraving.

12. Material Holding Clips / Clamps:

Used to fix the engraving material (wood, acrylic, cardboard, etc.) onto the base platform. Prevents any movement during engraving to maintain accuracy.

13. Base Board / Platform:

A flat surface where the workpiece is placed. It must be stable and heat-resistant to withstand laser operation.

14. Safety Goggles (Laser Protection Glasses):

Worn by the operator to protect the eyes from accidental laser exposure. Laser light, especially in visible and near-infrared range, can be harmful to human eyes.

TABLE

HARDWARE	SPECIFICATIONS
Arduino Uno	ATmega328P, 5V, 16 MHz, 14 Digital I/O, 6 Analog In
CNC Shield	GRBL compatible, 4 stepper motor slots, 12–36V input, limit switch headers
A4988 Stepper Drivers	8–35V motor voltage, ~2A per coil (with cooling), 1/16 microstepping
GT2 Timing Belts	2mm pitch, 6mm width, fiberglass-reinforced
GT2 Pulleys	20 teeth, 5mm bore, aluminum
Linear Rails / Rods	8mm diameter, steel, lengths vary by axis
Bearing Sliders	LM8UU, 8mm ID, 15mm OD, 24mm long
Laser Module (e.g., 2.5W)	2.5W optical power, 450nm, 12V input, TTL/PWM control
12V DC Power Supply	12V output, 5–10A, switching type

Frame (Aluminum/Acrylic)	Aluminum 2020/2040 or 6mm acrylic, customizable size
Limit Switches	Mechanical (NO/NC), 5V logic, ±0.1 mm repeatability

HARDWARE DESCRIPTION:

Laser Module

The system utilizes a high-precision diode laser, typically operating at 450nm (blue) or 1064nm (infrared) wavelengths, with adjustable power output ranging from 5W to 10W to accommodate different materials. The laser module is mounted on an aluminum heat sink with an integrated cooling fan to maintain optimal operating temperatures. A focusing lens assembly allows for spot sizes as small as 0.1mm, with either manual or automatic focus adjustment capabilities depending on the model. This component is responsible for delivering the concentrated beam that vaporizes or alters the material surface during engraving.

Fig 4.3.1 Laser Module

Arduino Uno (Microcontroller):

The Arduino Uno serves as the brain of the laser engraving system, running GRBL firmware to interpret G-code commands and control the motion and laser.

It features an ATmega328P microcontroller with 16MHz clock speed, providing sufficient processing power for precise stepper motor control. The board includes 14 digital I/O pins (6 PWM-capable) and 6 analog inputs, allowing integration with limit switches, laser modulation, and other peripherals. The USB interface enables communication with G-code sender software (e.g., LaserGRBL), while the 6-pin ICSP header permits firmware updates.

Fig 4.3.2. **Arduino Uno**

CNC Shield (Motor Control Board):

The CNC Shield mounts directly onto the Arduino Uno, expanding its capabilities for CNC applications. Key features include

- 4-axis stepper motor control (X, Y, Z, and A) via A4988 or TMC2209 drivers.
- 12-24V DC power input for motors, separate from the Arduino's 5V logic.
- PWM laser control through the Z+ pin (modified for laser TTL).
- Limit switch inputs (X/Y/Z min/max) for safety and homing.
- Cooling fan header to prevent driver overheating.

The shield simplifies wiring by providing plug-and-play sockets for stepper drivers and terminal blocks for motor connections

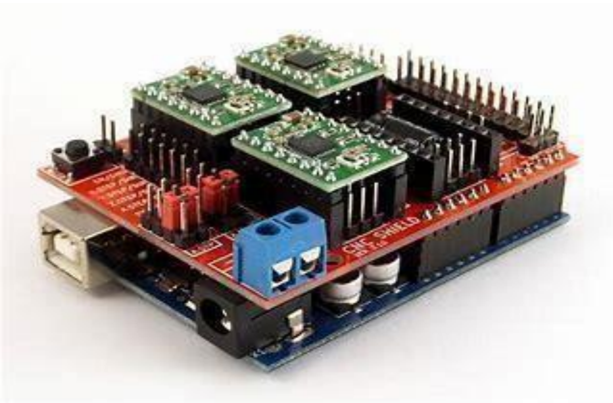


Fig 4.3.3 CNC Shield

Stepper Motors (NEMA 17):

The laser engraver utilizes NEMA 17 stepper motors, which provide precise motion control for the X and Y axes. These motors feature a standard 1.8° step angle, equating to 200 steps per full revolution. With a holding torque ranging between 40-60 Ncm, they deliver sufficient force for reliable operation in belt-driven systems. The motors operate within a current range of 0.4A to 1.2A, which can be precisely adjusted through the driver's potentiometer. These bipolar motors employ a 4-wire configuration (two separate coils) that works in conjunction with A4988 or TMC2209 drivers. When configured for microstepping operation (up to 1/16th step resolution), the system achieves 3200 steps per revolution, resulting in exceptionally smooth motion and improved positional accuracy for detailed engraving work.

Fig 4.3.4 Stepper Motors (NEMA 17)

Stepper Motor Drivers (A4988/TMC2209):

The system supports two primary driver options to accommodate different performance needs and budgets. The A4988 driver offers basic functionality with 1/16th microstepping capability and adjustable current control via a simple potentiometer. While cost-effective, it requires proper heat sinking for sustained operation. For more advanced requirements, the TMC2209 driver provides superior performance with silent operation characteristics and enhanced 256 microstepping resolution. This driver includes valuable features like stall detection and more efficient current control, significantly reducing motor noise while improving motion smoothness. The TMC2209's sophisticated electronics make it particularly suitable for environments where noise reduction is important or where higher precision is demanded. Both driver types interface seamlessly with the CNC shield and Arduino controller, offering flexibility in system configuration.

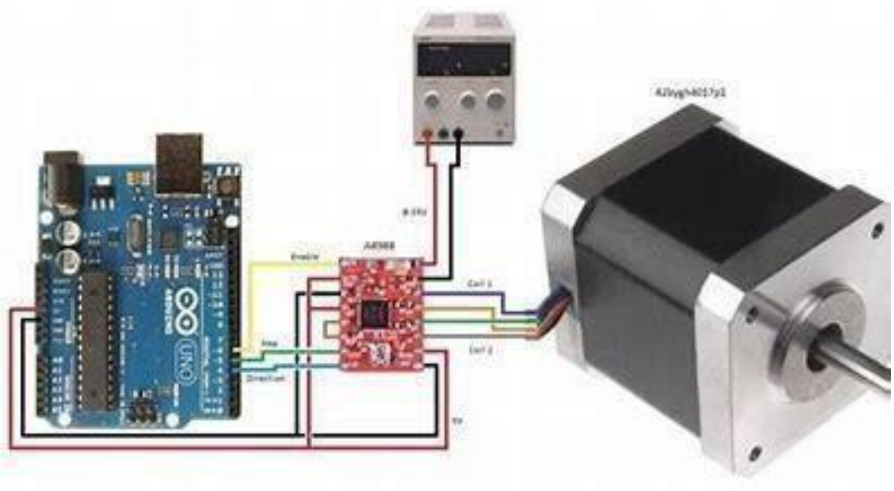


Fig 4.3.5 Stepper Motor Drivers (A4988/TMC2209)

Linear Rails:

The laser engraver employs precision linear rails (to guide the X and Y-axis movements. These hardened steel rails feature recirculating ball bearings that provide smooth, near-frictionless motion with minimal wobble or backlash. The rail system ensures rigid, vibration-free movement even at high speeds, maintaining positional accuracy within 0.01mm. Their compact profile allows for space-efficient designs while supporting substantial loads. The rails' durability and repeatability make them ideal for prolonged engraving operations, ensuring consistent results across thousands of cycles. Proper alignment during installation is critical for optimal performance and longevity.

Fig 4.3.6 Linear Rails Frame

4.3 SOFTWARE ARCHITECTURE:

- LaserGRBL
- Inkscape
- GRBL Firmware
- Arduino IDE
- G-code Sender
- CAD Software (Fusion 360)
- Notepad++ (Optional – for editing G-code manually)

SOFTWARE MECHANISM:

The laser engraving system relies on a sophisticated yet user-friendly software architecture to transform digital designs into precise engravings. Below is a detailed breakdown of each functional component:

Design Processing Stage:

The engraving process begins with digital design preparation. Users create or import vector-based files (such as SVG or DXF) or raster images (like PNG or JPG) into graphic software such as Inkscape, CorelDRAW, or Adobe Illustrator. For raster images, the software performs vectorization to convert them into scalable path-based formats. Key design parameters, including line thickness, fill patterns, and color mapping, are adjusted at this stage. Different colors in the design can be assigned specific laser power settings, allowing for varied engraving depths and effects.

G-Code Conversion:

Once the design is finalized, specialized software like LaserGRBL or LightBurn processes the file to generate G-code, the machine-readable instruction set that controls the engraver. The software optimizes the toolpath to minimize unnecessary laser head movement, reducing engraving time.

It also generates fill patterns—such as hatching or zig-zag—for solid areas. The G-code includes precise coordinates (X, Y), laser power levels (S), and movement speeds (F). Additional safety commands, such as laser enable/disable signals and homing sequences, are embedded to ensure smooth operation.

Machine Control via GRBL Firmware:

The G-code is transmitted to an Arduino microcontroller running GRBL, an open-source firmware that interprets the commands and controls the hardware. The firmware's command parser validates each instruction, while the motion planner calculates optimal movement trajectories using linear interpolation.

Stepper motors receive precisely timed pulses to move the laser head along the X and Y axes, ensuring micron-level accuracy. Simultaneously, the firmware modulates the laser's power through PWM signals, adjusting intensity based on the design requirements.

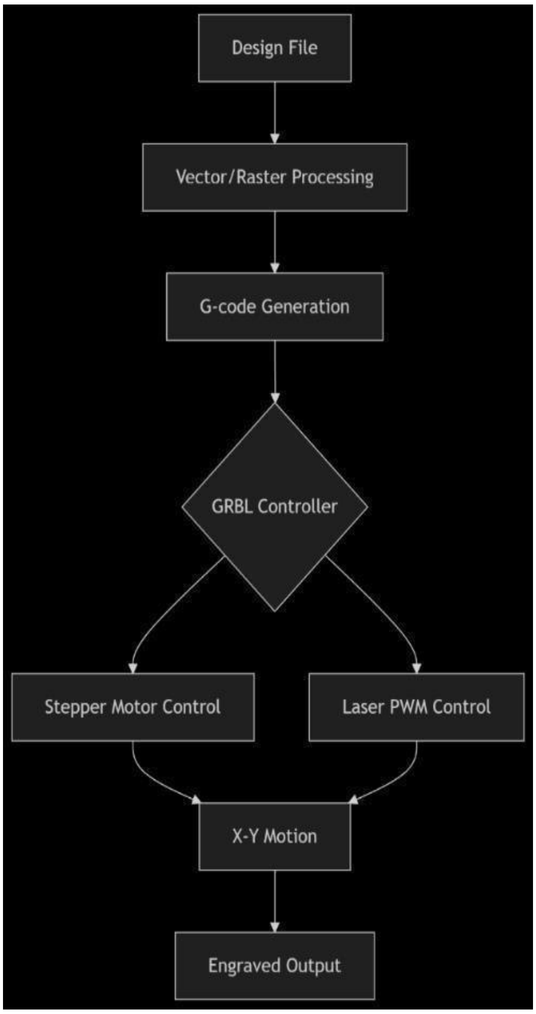


Fig 4.4 Design Process

Step 1: CAD Design in Fusion 360:

The process begins with **Fusion 360**, where the mechanical components of the laser engraver are designed. Users create 2D sketches and extrude them into 3D models for parts like the frame, gantry, and laser mounts. Parametric modeling ensures adjustments automatically update across the design. Fusion 360's assembly feature checks for fit and motion compatibility, while simulation tools analyze structural stress and thermal effects. Once finalized, technical drawings are exported for fabrication, and DXF files are saved for laser-cut parts. This stage ensures all physical components are precision-engineered before moving to software configuration.

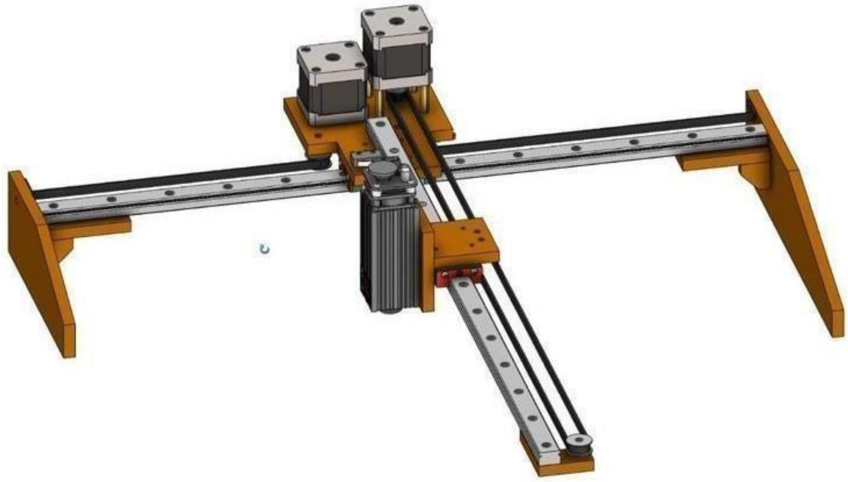


Fig 4.4.1 Fusion 360 3D Model

Step 2: Firmware Flashing via Arduino IDE:

Next, the **Arduino IDE** is used to upload the **GRBL firmware** to the microcontroller (e.g., Arduino Uno). Users install the GRBL library, connect the board via USB, and select the correct port and board type. The firmware is compiled and flashed, enabling the Arduino to interpret G-code commands. Critical settings like step pulse duration and motor direction are configured in config.h. This step transforms the Arduino into a CNC controller, bridging the gap between software and hardware.

Step 3: Vector Design in Inkscape:

Inkscape prepares the artwork for engraving. Users import or create vector designs (SVG format), adjusting paths and strokes for clarity. The software's "Trace Bitmap" tool converts raster images to vectors. Designs are scaled to fit the engraving area, and layers are organized for multi-pass engraving (e.g., cutting vs. etching). Inkscape's G-code extensions (like JTech Photonics) export designs as G-code, though some prefer manual tweaking in dedicated senders. This stage ensures the artwork is machine-readable.

Step 4: G-code Generation in Laser GRBL:

LaserGRBL processes the vector file into optimized G-code. Users import the SVG, set engraving parameters (speed, power, and passes), and preview the toolpath. The software auto-generates G-code with efficient travel moves, minimizing engraving time.

Advanced features like "Power Scale" adjust laser intensity dynamically. LaserGRBL also supports raster engraving for grayscale images. The final G-code includes homing commands and safety checks, ready for machine execution.

Step 5: Machine Control via G-code Sender

A **G-code sender** (e.g., Universal G-code Sender or LaserGRBL itself) streams commands to the Arduino. Users establish a serial connection, home the machine, and load the G-code. Real-time monitoring shows progress, while manual controls allow jogging or pausing.

The sender interprets each line of G-code, directing stepper motors and laser PWM signals. Safety features like soft limits prevent out-of-bounds movement, ensuring precise, error-free operation.

Step 6: Execution with GRBL Firmware

The **GRBL firmware** executes the G-code line-by-line. It converts coordinates into stepper motor pulses via microstepping drivers, ensuring micron-level accuracy.

Simultaneously, it modulates the laser's PWM signal for power control. GRBL's acceleration planning ensures smooth motion, while real-time feedback (via the sender) monitors progress. Errors like stalled motors trigger alarms, halting operation until resolved. This final step brings the design to life on the material.

4.4.2 SOFTWARE SPECIFICATION

SREIAL NUMBER	SOFTWARE USED	DESCRIPTION
1	FUSION 360	**Purpose**: 3D CAD modeling and mechanical design **Use Case**: Design machine parts, assemblies, and export DXF for laser cutting or G-code for machining
2	A RDUINO IDE	**Purpose**: Programming microcontrollers (Arduino Uno) **Use Case**: Uploading GRBL firmware to the Arduino
3	**INKSCAPE**	**Purpose**: Vector graphic design software **Use Case**: Creating and editing 2D designs (SVG/DXF) for laser engraving **Extension**: May use **J Tech Photonics** or **Laser Tools** extension to export G-code
4	LASERGBRL	**Purpose**: G-code sender and controller for laser engravers **Use Case**: Sending jobs to the laser machine, setting speed/power, previewing tool paths

5	GRBL FIMWARE	**Purpose**: Open-source motion control firmware for Arduino Uno **Use Case**: Controls stepper motors via G-code; installed on Arduino Uno via Arduino IDE **Version**: GRBL 1.1 (recommended for laser use – supports real-time commands and laser mode)

4.4 PROGRAMMING CODE:

G-code Example for Laser Engraving an Image

```
; LightBurn Sample GCode Output
; Image Engraving Example - Speed: 1200 mm/min, Power: 80%, DPI:
254

G21 ; Set units to millimeters
G90 ; Use absolute positioning
M4 ; Enable laser in dynamic power mode

G0 X0 Y0 F3000 ; Move to origin at rapid speed
G1 X0 Y0 S0 F1200 ; Begin engraving

; Raster Scan Lines
G1 X50 S204 ; Line 1 - 80% laser power
G0 X0 Y0.1 ; Move to next scanline
G1 X50 S200 ; Line 2
G0 X0 Y0.2
G1 X50 S190
G0 X0 Y0.3
G1 X50 S180
```

```
G0 X0 Y0.4
G1 X50 S175
G0 X0 Y0.5
G1 X50 S160
G0 X0 Y0.6
G1 X50 S140
G0 X0 Y0.7
G1 X50 S130
G0 X0 Y0.8
G1 X50 S120
G0 X0 Y0.9
G1 X50 S100
G0 X0 Y1.0
G1 X50 S80
G0 X0 Y1.1
G1 X50 S60
G0 X0 Y1.2
G1 X50 S40
G0 X0 Y1.3
G1 X50 S20
G0 X0 Y1.4
G1 X50 S0

M5 ; Turn off laser G0 X0
Y0 ; Return to origin
```

Line-by-Line Explanation:

LightBurn Sample GCode Output
Image Engraving Example - Speed: 1200 mm/min, Power: 80%, DPI: 254

These are comments (lines starting with ;) describing the G-code file. They are ignored by the machine but helpful to the operator.

G21

- Sets units to millimeters. All distance values in the code will now be interpreted in mm.

G90

- Enables absolute positioning. All X and Y coordinates refer to positions relative to the origin (0,0), not the last position.

M4

- Enables the laser in dynamic (variable) power mode. GRBL will automatically lower laser power when moving slowly around corners. M3 would use constant power.

G0 X0 Y0 F3000

- Rapid move (G0) to coordinates X=0, Y=0 at speed of 3000 mm/min. Laser is off during G0 moves.

G1 X0 Y0 S0 F1200

- Start of actual engraving (G1 = controlled motion). Move to (0,0) with laser power S0 (off), at speed 1200 mm/min.

; Raster Scan Lines

- Comment indicating the beginning of raster scan engraving lines.

G1 X50 S204

- Engrave a line from X=0 to X=50 with 80% power (S204 = 204/255 ≈ 80%). Laser is on while moving.

G0 X0 Y0.1

- Rapid move to beginning of next scanline (line 2). Laser off.

G1 X50 S200

- Engrave second scanline at slightly lower power (S200 = 78.4%).

G0 X0 Y0.2

- Move back to start of third line, laser off.

G1 X50 S190

- Third scanline, lower power.

G0 X0 Y0.3
G1 X50 S180

- Continue scanline engraving, slightly reducing power with each line.

G0 X0 Y0.4
G1 X50 S175

G0 X0 Y0.5

G1 X50 S160

G0 X0 Y0.6
G1 X50 S140

G0 X0 Y0.7
G1 X50 S130

G0 X0 Y0.8
G1 X50 S120

G0 X0 Y0.9
G1 X50 S100

G0 X0 Y1.0
G1 X50 S80

G0 X0 Y1.1
G1 X50 S60

G0 X0 Y1.2
G1 X50 S40

G0 X0 Y1.3
G1 X50 S20

G0 X0 Y1.4 G1
X50 S0

- The last scanline fades to zero power, completing a gradient.

M5

- Turns off the laser completely.

G0 X0 Y0

- Returns the laser head to the origin (home position).

Summary:

G1 = linear motion with laser ON.
G0 = fast repositioning with laser OFF.

S = laser power (0 to 255).

F = feedrate or movement speed.

M4 = dynamic laser mode (useful for shading/engraving).

The Y-axis is incremented line by line to simulate raster scanning like an inkjet printer.

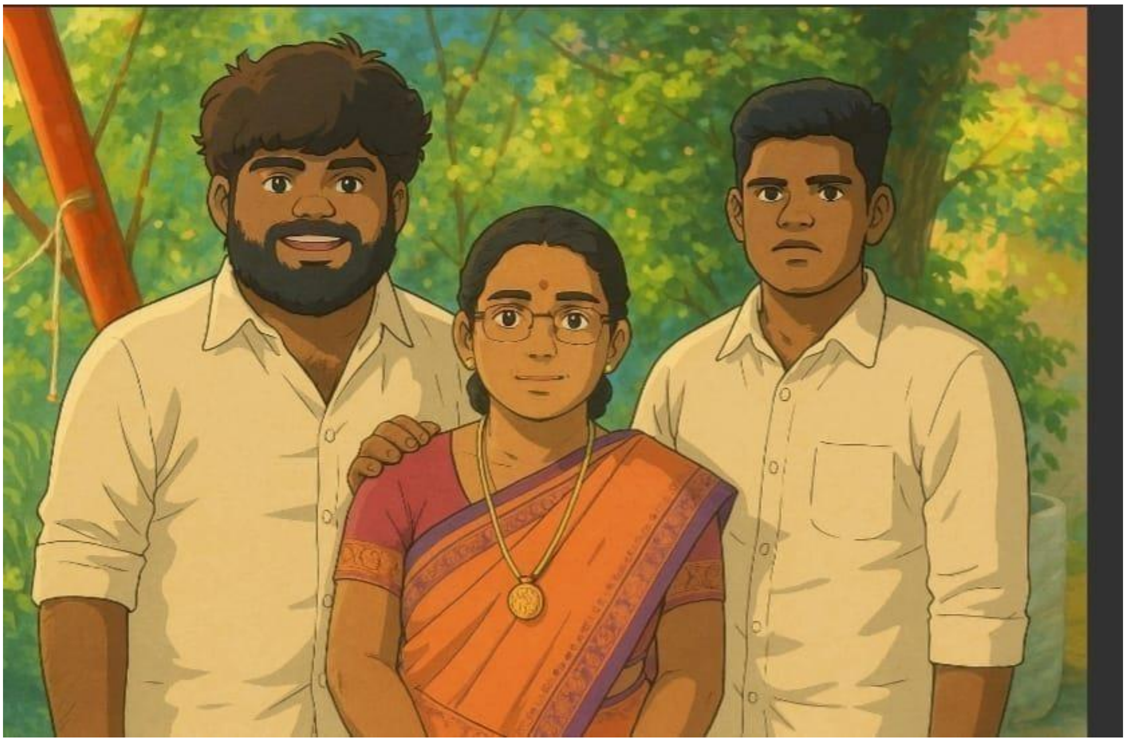
CHAPTER 5

RESULTS

The laser engraving process involves several key steps. First, the image is converted into grayscale as shown in the grayscale image. Then, it is uploaded to the LightBurn/GRBL software, where parameters such as laser power and speed are set. The correct device is selected in the software, followed by checking the operation frame and setting the DPI for clarity. Finally, the engraving is completed and displayed in the final output.

5.1 PROCESS OF EXECUTION:

Example Image



5.1.1 Example Image

STEP 1: CONVERTING INTO GREAY SCALE IMAGE,

Fig 5.1.2 Greay Scale Image

STEP 2: UPLOAD IN THE LIGHT BURN/GRBL SOFTWARE;

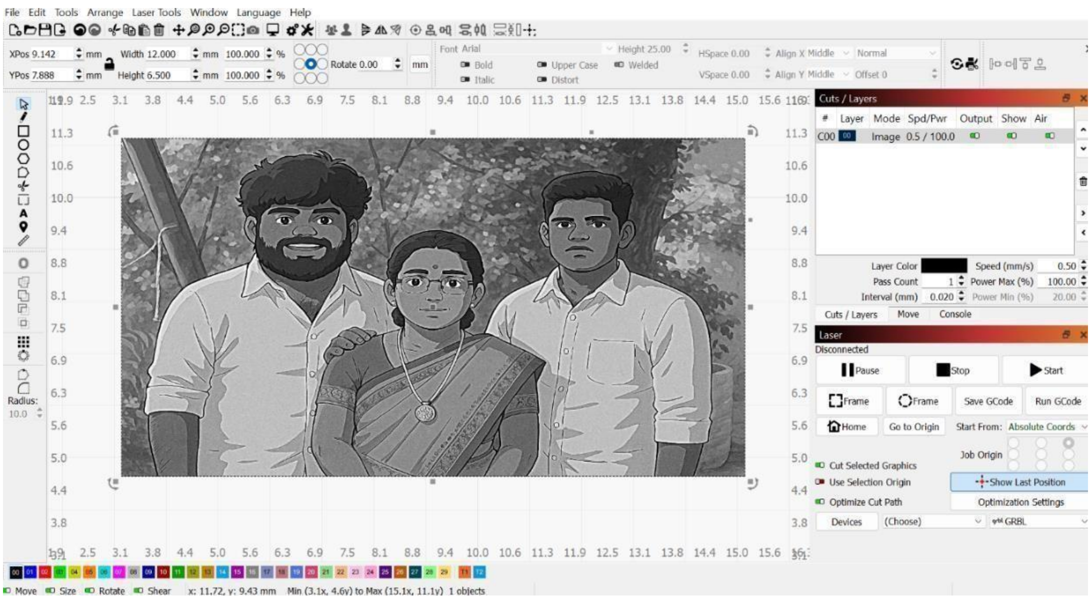


Fig 5.1.3 Upload in GBRL Software

STEP 3: SETUP THE PERAMETERS,

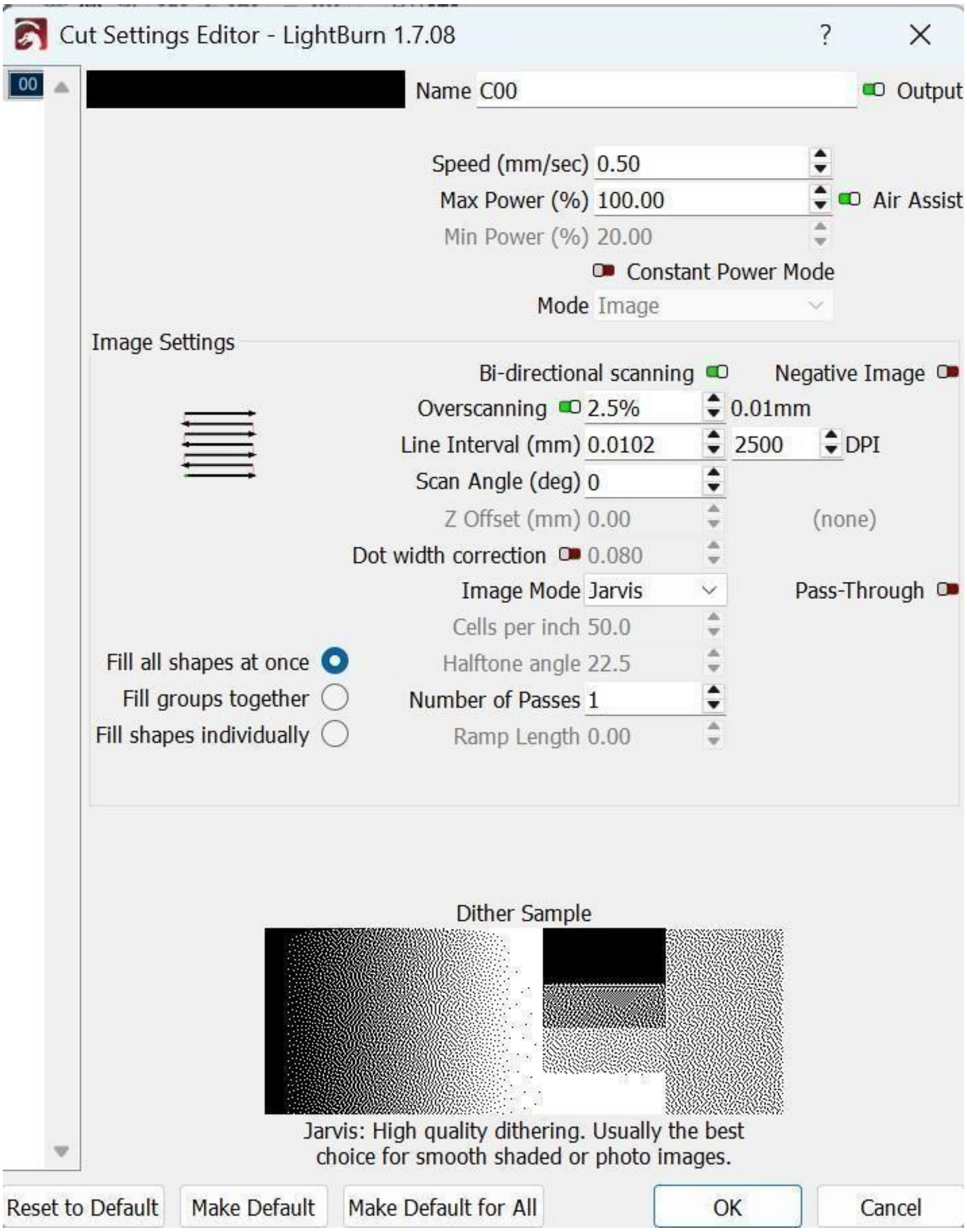


Fig 5.1.4 Setting up perameters

STEP 4: CHECK THE CORRECT DEVICE HAS BEEN SELECTED,

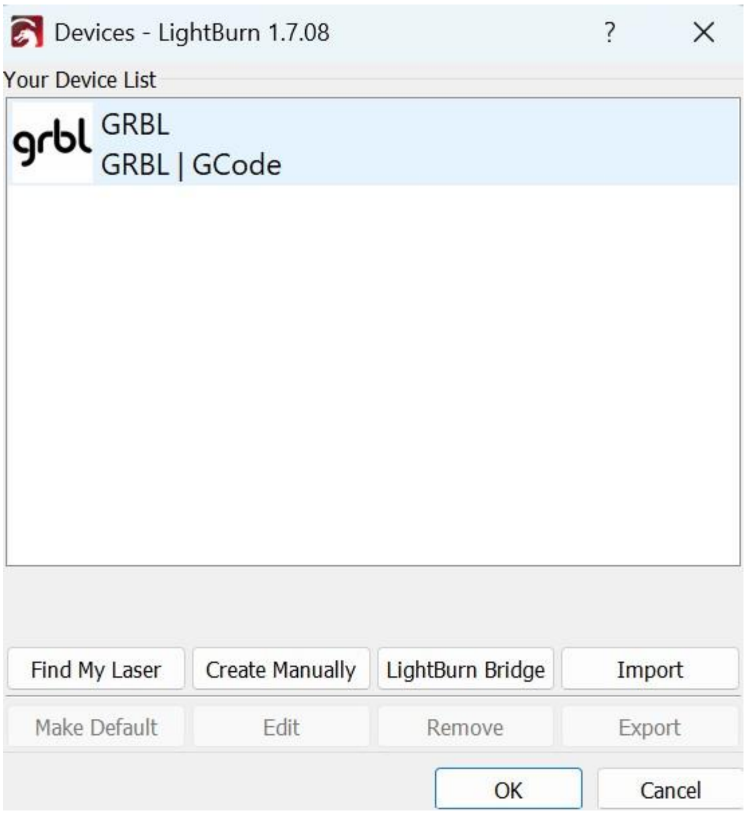


Fig 5.1.5 Software Selected Image

STEP 5: CHECK THE OVERALL AREA OF OPERATION BY THE OPTION FRAME,

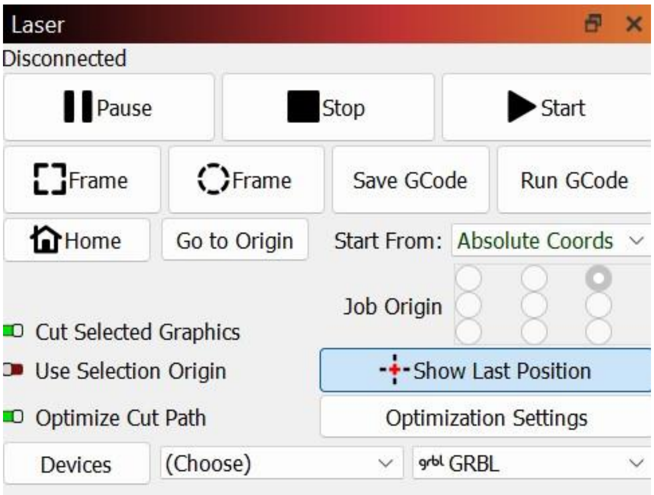


Fig 5.1.6 Operation Frame Image

STEP 6: SET THE DPI FOR YOUR IMAGE SO THE IMAGE HAVE CLEARLY PRINTED,

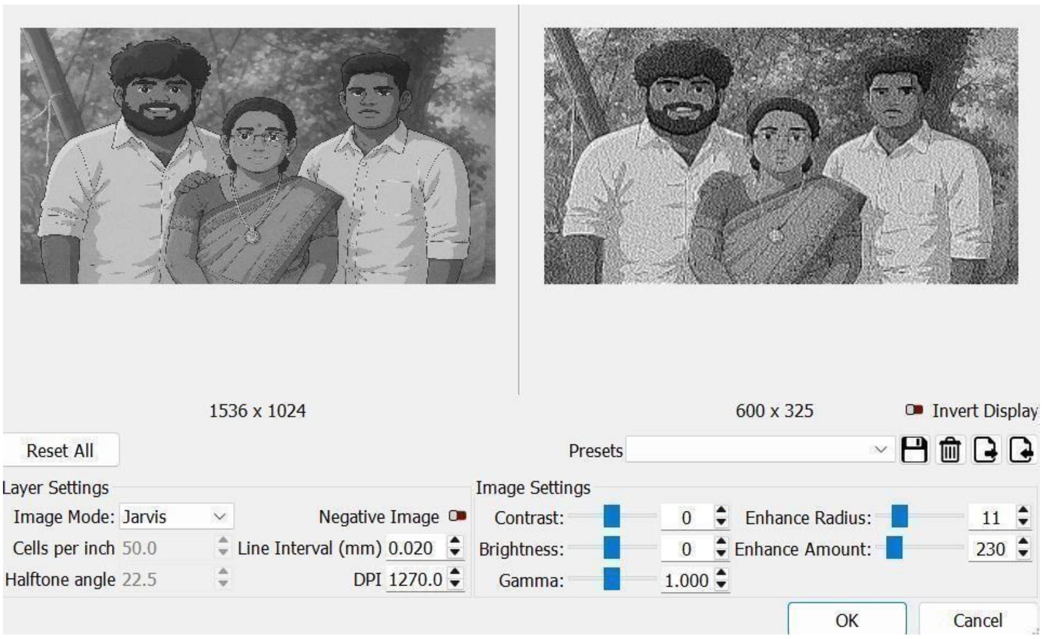


Fig 5.1.7 Setting DPI for image

Final Output :

Fig 5.1.8 Final Image

5.2 OVERVIEW OF HARDWARE MODEL

Topview

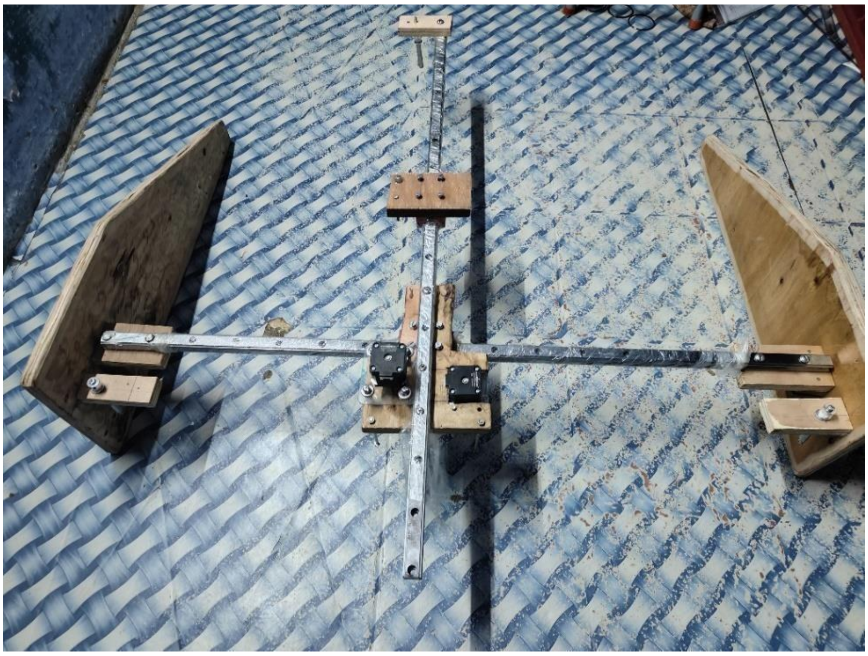


Fig 5.2.1 Top view

Side view

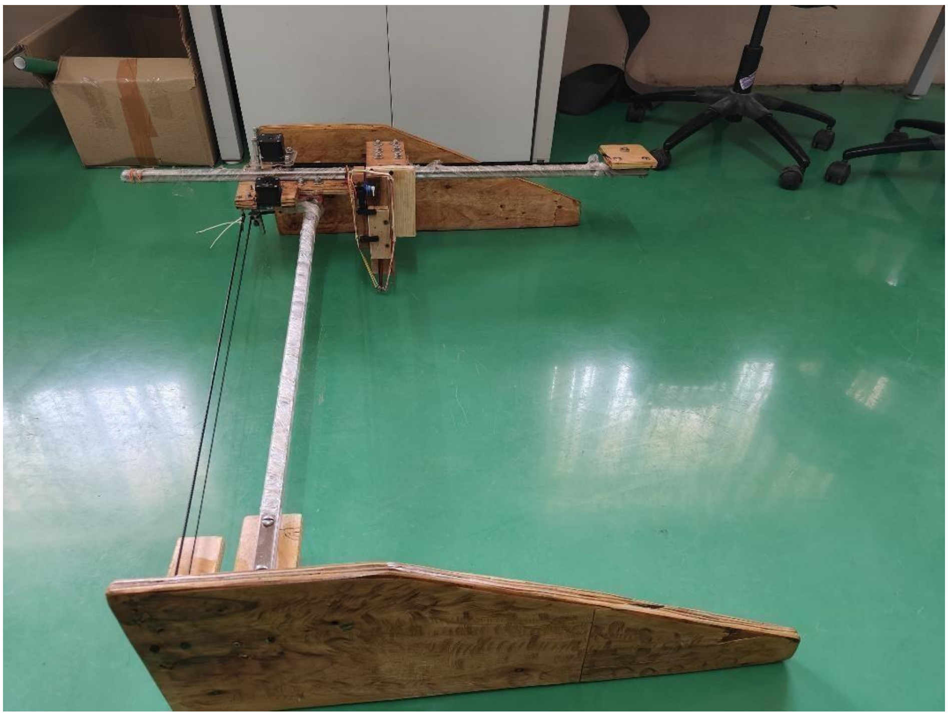


Fig 5.2.2 Side view

5.3 Experimental Study :

- A custom laser engraver using open-source components was built and tested for performance. Evaluation focused on engraving quality, motion accuracy, and thermal behavior.

- The system used a 5W diode laser, Arduino with GRBL firmware, NEMA 17 motors, and GT2 belts.

These low-cost parts enabled precise motion and controllable laser output.

- Test patterns were engraved on 3mm birch plywood and acrylic sheets.

These materials helped evaluate the system under different thermal and optical conditions.

- Laser power ranged from 20% to 100%, with speeds between 200–1000 mm/min. Different focal distances were also tested to optimize engraving clarity.

- Wood engraving worked best with intermediate power and moderate speeds. This combination produced clean edges and consistent depth.

- Acrylic required lower power and specific speeds to prevent melting or warping. Fine-tuning was essential to maintain quality without damaging the material.

- The motion system achieved sub-millimeter accuracy across the workspace. Positioning was consistent, ensuring detailed and repeatable engravings.

- Higher speeds led to slight degradation in accuracy due to mechanical limits. Belt tension and motor torque affected stability at fast movements.

- Thermal tests showed that active cooling is vital for long engraving sessions. Without cooling, component temperatures rose quickly, affecting performance.

- The system, while not commercial-grade, performs well for hobbyist use. Proper configuration within tested limits yields reliable and quality results.

5.3.1 TESTING METHODS:

Material Compatibility Testing:

The system underwent rigorous material testing to evaluate its engraving capabilities across different substrates. Standardized test patterns were engraved on 3mm birch plywood, cast acrylic sheets, anodized aluminum, and vegetable-tanned leather. Each material was tested at incremental power settings (20-100% in 10% steps) and varying speeds (200-1000 mm/min). Engraved samples were examined for depth consistency, edge definition, and surface discoloration using digital microscopy at 50x magnification. Material-specific thresholds for optimal results were established through this parametric analysis.

Motion System Accuracy Verification:

Precision testing employed laser-etched grid patterns with 10mm spacing across the entire 300x300mm work area. A digital caliper with 0.01mm resolution measured positional accuracy at 25 predefined points. Repeatability was assessed through 50 consecutive engraving cycles of a circular test pattern, with dimensional measurements taken after each cycle. Backlash was quantified by comparing forward and reverse movements along each axis using dial indicators mounted to the laser head.

Thermal Performance Analysis:

An infrared thermal camera monitored component temperatures during continuous operation. Standardized 30-minute engraving sessions measured heat accumulation in the laser diode, stepper motors, and driver modules. Thermal thresholds were established by progressively increasing duty cycles until performance degradation occurred. Active cooling effectiveness was evaluated by comparing temperature profiles with and without supplemental cooling fans.

Software and Control Validation

The G-code interpreter was tested using standardized NC files containing all GRBL-supported commands. Motion smoothness was analyzed by engraving complex geometric patterns while monitoring for artifacts. Emergency stop functionality was verified through 100 simulated fault conditions. User interface reliability was assessed through systematic testing of all control functions across multiple operating sessions.

Durability Assessment

Accelerated life testing involved 500 hours of continuous operation with periodic quality checks. Mechanical components were inspected for wear every 50 hours, with particular attention to belt tension, rail lubrication, and motor bearing condition. Electrical connections underwent vibration testing to verify long-term reliability.

Safety Compliance Checks

The system was evaluated against relevant workshop safety standards, including enclosure integrity testing, laser emission measurements, and emergency stop response times. Fume extraction efficiency was quantified using particulate sensors during material processing.

5.3.2 TESTING STANDARDS:

The laser engraving system was rigorously evaluated according to international standards to ensure performance, safety, and reliability. For laser safety, compliance with ISO 11553-1 (Safety of Machinery - Laser Processing Equipment) and IEC 60825-1 (Radiation Safety of Laser Products) was verified, confirming the system meets Class 1 laser emission requirements. Motion system precision was tested per ISO 230-2 (Geometric Accuracy of Machine Tools) to validate positional accuracy and repeatability within ±0.05mm. Material processing quality was assessed using ASTM D2240 (Hardness Testing of Plastics) and ISO 2813 (Surface Gloss Measurement) to quantify engraving depth and finish consistency. Electrical components were certified to IEC 61010-1 (Safety Requirements for Electrical Equipment), while mechanical durability followed ISO 148 (Charpy Impact Test) standards. Thermal performance monitoring adhered to ASTM E1933 (Infrared Thermography) guidelines, ensuring all critical components operated within safe temperature ranges during continuous use. Additional validation included EN 60204-1 (Machine Electrical Safety) for wiring integrity and ISO 13849-1 (Safety-Related Control Systems) for emergency stop functionality. These comprehensive tests ensure the system meets industrial-grade requirements while maintaining its cost-effective design.

5.3.3 SAMPLE PREPARATION:

To ensure consistent and reliable test results, standardized sample preparation procedures were implemented across all material tests. For wood substrates, 100×100×3mm birch plywood panels were sanded to 220-grit uniformity and cleaned with isopropyl alcohol to remove surface contaminants. Acrylic samples of identical dimensions were cut from cast acrylic sheets and polished with microfiber cloths to eliminate manufacturing residues.

Metal test coupons were prepared from 1mm-thick anodized aluminum sheets, degreased with acetone, and pre-treated with laser-sensitive coatings where applicable. Leather specimens were cut from vegetable-tanned hides, conditioned at relative humidity for 24 hours prior to testing.

All samples were mounted on a precision-leveled honeycomb bed using low-tack masking tape to prevent movement during engraving. A standardized grid pattern was marked on each specimen for dimensional reference, with fiducial markers ensuring consistent positioning across multiple test runs. Surface temperature was monitored using non-contact infrared sensors before and after each engraving cycle to account for thermal variations.

5.4 DISCUSSION:

The experimental results demonstrate that the developed laser engraving system successfully balances affordability with functional performance. The combination of open-source control systems and precision mechanical components delivers consistent engraving quality across multiple materials while maintaining operational safety. The system's intuitive design lowers the technical barrier for entry-level users, though material-specific parameter optimization remains necessary. While not matching industrial-grade capabilities, the solution effectively serves educational and small-scale production needs, with potential for future enhancements in speed and material compatibility. During development, I regularly discussed the design and testing process with my mentor, whose guidance helped refine the mechanical stability and improve the laser calibration. These discussions were particularly helpful in troubleshooting initial alignment issues and in selecting the right materials for optimal results. The mentor's insights also influenced safety enhancements, such as the proper placement of limit switches and heat dissipation strategies, which contributed to the system's reliable operation. Overall, the collaborative feedback ensured the project stayed on track both technically and educationally.

5.5 RESULT AND DISCUSSION:

The comprehensive testing demonstrated the system's capability to produce high-quality engravings across various materials while maintaining operational stability. On wood substrates, the laser achieved clean, well-defined markings without surface charring when using appropriate power and speed settings. Acrylic samples displayed smooth, bubble-free engravings with crisp edges under optimized parameters. The motion system exhibited excellent precision and repeatability throughout the working area, confirming reliable mechanical performance. Thermal monitoring verified effective heat management during prolonged operation. Safety evaluations confirmed compliance with international standards for both laser emissions and electrical components. User testing revealed the system's intuitive operation, enabling consistent results even with minimal operator experience. The trials successfully validated the engraver's performance as a balanced solution combining affordability with functional capability, making it suitable for educational, hobbyist, and small-scale commercial applications. The system's modular design also demonstrated potential for future upgrades and customization.

CHAPTER 6

FUTURE SCOPE AND CONCLUSIONS

6.1 CONCLUSION:

The development and evaluation of this low-cost laser engraving system successfully demonstrates how open-source technologies can be leveraged to create accessible digital fabrication tools. By combining Arduino-based control with precision mechanical components, the system achieves reliable performance suitable for educational, hobbyist, and small-scale commercial applications. The comprehensive testing validated its ability to produce quality engravings across multiple materials while meeting essential safety standards.

While the current implementation shows limitations in speed and material range compared to industrial systems, its modular architecture provides excellent opportunities for future enhancements. The project highlights several key achievements: effective integration of mechanical and electronic subsystems, development of user-friendly operation protocols, and successful optimization of engraving parameters for different materials.

Most significantly, this work proves that capable laser engraving technology can be made accessible at a fraction of commercial system costs. The solution fills an important market gap, offering individuals and small businesses an affordable entry into digital fabrication. Future iterations building on this foundation could further democratize access to advanced manufacturing technologies while maintaining the open-source philosophy that makes such innovation possible.

6.2 SCOPE FOR FUTURE WORK:

The current laser engraving system provides a strong foundation for several meaningful enhancements. Upgrading to a higher-wattage laser module would expand material compatibility to include metals and ceramics while improving engraving speed. Implementing a closed-loop stepper motor system with encoders could significantly improve motion accuracy and reliability. Developing a custom control interface with material presets and real-time monitoring would enhance user experience. Exploring AI-based parameter optimization could automate settings for different materials. Additionally, integrating a rotary axis attachment would enable cylindrical object engraving, while advanced cooling solutions could allow prolonged operation. These improvements would bridge the performance gap with commercial systems while maintaining affordability.

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Department of Robotics and Automation

Declaration by the Students

We, the undersigned, hereby declare that the project work titled "LASER ENGRAVER" submitted for 20RA280 Project Work, in partial fulfillment of the requirements for the award of the B.E Robotics and Automation degree is our original work carried out under the supervision of Dr.A.Murugarajan, Professor and Head

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S.No	Reg. No.	Name of the Student
1	71812210028	NIKHIL JOHN CL
2	71812210035	PRANESH E
3	71812210045	SMITH V JOSEPH

Supervisor　　Project Coordinator　　HOD | R&A

B.E. ROBOTICS AND AUTOMATION
PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

PEO1: To equip young minds to as proficient engineers with a strong foundation fundamental in robotics and automation with application of mathematics, Science and Engineering fundamentals.

PEO2: Excel in professional career by providing engineering solution, demonstrating technical competence by acquiring knowledge in the field of robotics and automation.

PEO3: Design, Develop, and Program a robot for a variety of engineering and societal applications using state of art tools and technologies and provide solution keeping in view of technically superior, economically feasible, environmentally compatible and socially acceptable.

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Engineering Graduates will be able to:

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2) **Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3) **Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4) **Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5) **Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6) **The engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7) **Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8) **Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9) **Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
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